Kinematics 1D and 2D Revision Sheet

Level 1 Sheet **Maximum Time Taken to Solve: 60 Min**

Note: Revise Class Notes + Do Last 5 Years of PYQ + These Sheets of Level 1 to 4.

1. A juggler keeps on moving four balls in air continuously such that each ball attains 20 m height. Time interval between any two projections is same. When the first ball leaves his hand, the position of other balls (in metre) will be: (take $g = 10\ \text{m/s}^2$)

(1) 15, 20, 15 (2) 10, 20, 10 (3) 5, 15, 20 (4) 5, 10, 20

2. A person travelling on a straight line moves with a uniform velocity v_1 for some distance and with uniform velocity v_2 for the next equal distance. The average velocity v is given by

(1) $v = \dfrac{v_1 + v_2}{2}$ (2) $v = \sqrt{v_1 v_2}$ (3) $\dfrac{2}{v} = \dfrac{1}{v_1} + \dfrac{1}{v_2}$ (4) $\dfrac{1}{v} = \dfrac{1}{v_1} + \dfrac{1}{v_2}$

3. A particle starts from rest and moves along a straight line with acceleration given by

$$a = \begin{cases} (1 + 2t)\ \text{m/s}^2, & 0 \le t \le 5 \\ 3\ \text{m/s}^2, & t > 5 \end{cases}$$

then the velocity (in m/s) of the particle at $t = 7$ sec is :

(1) 12 (2) 24 (3) 36 (4) 48

4. A body dropped from top of a tower, covers a distance $7h$ in last second of it's motion, where h is distance covered in first second, then total time of flight will be.

(1) 2 s (2) 3 s (3) 4 s (4) 8 s

5. A car moving with a speed of 12 m/s can be stopped by applying brakes after travelling a distance of 6 m. If the same car is moving with a speed of 24 m/s then the stopping distance of the car is :

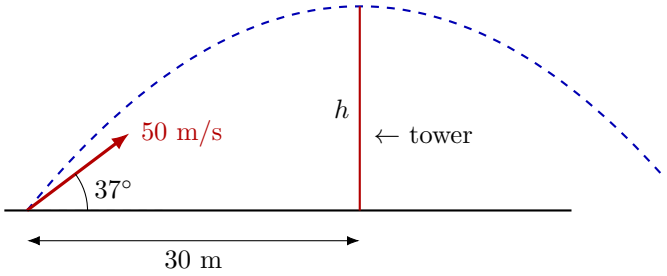
(a) 24 m (b) 12 m (c) 18 m (d) 30 m

6. A balloon is ascending at the rate of $9.8\ \text{m s}^{-1}$ at a height of 39.2 m above the ground when a food packet is dropped from the balloon. After how much time and with what velocity does it reach the ground? Take $g = 9.8\ \text{m s}^{-2}$.

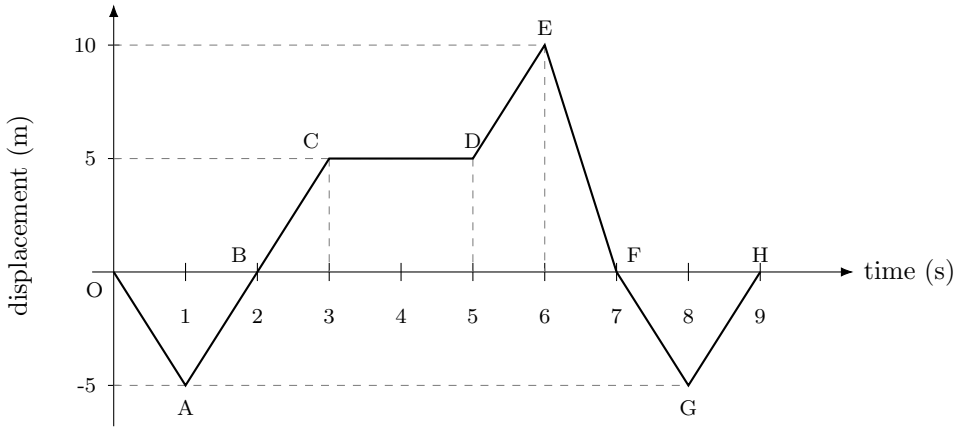
7. A particle is thrown upward from ground with speed 'u' if displacement of particle in 2nd & 7th sec is same, find 'u' ?

8. A person travelling on a straight line moves with a uniform velocity v_1 for a distance x and with a uniform velocity v_2 for the next $\dfrac{3}{2}x$ distance. The average velocity in this motion is $\dfrac{50}{7}$ m/s. If v_1 is 5 m/s then $v_2 =$ _____ m/s.

9. Find value of 'h' if projectile is grazing the tower ?



10. The displacement x versus time graph is shown below :



(A) The average velocity during 0 to 3 s is 10 m/s

(B) The average velocity during 3 to 5 s is 0 m/s

(C) The instantaneous velocity at $t = 2$ s is 5 m/s

(D) The average velocity during 5 to 7 s and instantaneous velocity at $t = 6.5$ s are equal

(E) The average velocity from $t = 0$ to $t = 9$ s is zero

Choose the correct answer from the options given below :

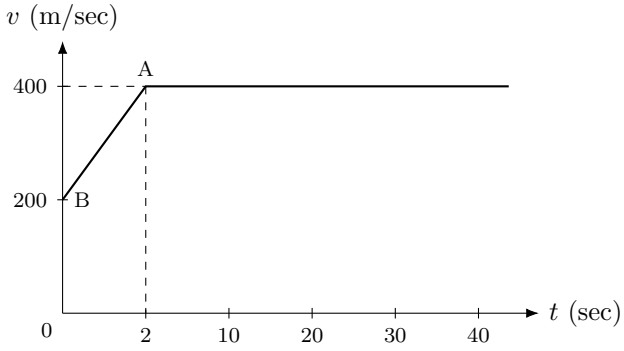
(a) (A), (D), (E) only

(b) (B), (C), (D) only

(c) (B), (D), (E) only

(d) (B), (C), (E) only

11. The motion of an airplane is represented by velocity-time graph as shown below. The distance covered by airplane in the first 30.5 second is _____ km.



(a) 12 (b) 3 (c) 6 (d) 9

12. A gas balloon is going up with a constant velocity of 10 m/s. When this balloon reached a height of 75 m, a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is _____ m. (Take $g = 10$ m/s^2)

(A) 85 (B) 150

(C) 129 (D) 125

13. From 18 m height above the ground a ball is dropped from rest. The height above the ground at which the magnitude of velocity equal to the magnitude of acceleration (in the same set of units) due to gravity is _____ m.
(Take $g = 10$ m/s^2 and neglect the air resistance)

14. Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground, at the instant when the first drop strikes the ground is _____ m. ($g = 10$ m/s^2)

(A) 4.2 (B) 2.5

(C) 4.0 (D) 3.8

15. A helicopter flying horizontally with a speed of 360 km/h at an altitude of 2 km, drops an object at an instant. The object hits the ground at a point O, 20 s after it is dropped. Displacement of 'O' from the position of helicopter where the object was released is :
(use acceleration due to gravity $g = 10$ m/s^2 and neglect air resistance)

(A) $2\sqrt{5}$ km (B) 4 km

(C) 7.2 km (D) $2\sqrt{2}$ km

16. A particle is moving in a straight line. The variation of position x as a function of time t is given as $x = (t^3 - 6t^2 + 20t + 15)$ m. The velocity of the body when its acceleration becomes zero is:

(A) 4 m s^{-1} (B) 8 m s^{-1}

(C) 10 m s^{-1} (D) 6 m s^{-1}

Keep Going!

"A journey of a thousand miles begins with a single step."

— Lao Tzu